Semester DB Project: Phase II

CSC 436 | Review, Feedback and Revisions

Part I: Review Feedback and Revisions

1. Summary of Feedback Received

Our group carefully reviewed all feedback provided during the Phase II presentations and peer reviews. The overall response was positive, with reviewers acknowledging the strength of our database design, our consideration for future scalability with programming languages, and the normalization work we completed prior to submission.

The feedback contained two categories of comments. The first was a recognition of our existing 3NF compliance. The second was forward looking advice about optional improvements that could benefit the database in future development, not corrections to existing errors. No feedback identified a violation of normalization rules, broken constraints, or incorrect query logic in our current implementation.

2. Revisions and Weaknesses Addressed

After thoroughly reviewing the feedback as a group, we determined that no structural revisions to the database are required at this time. Our reasoning for each point of feedback is documented below.

Feedback Point 1: Composite Primary Keys on Junction Tables

The feedback suggested that our junction tables, function_implementation, operator_implementation, and structure_implementation, could use composite primary keys combining implementation_id with the linked entity ID. This was presented as a scalability consideration for future many to many expansion, not as a flaw in our current design.

Our current implementation correctly enforces referential integrity through foreign key constraints on both columns in each junction table. The existing structure functions as intended for the current scope of the project. Implementing composite keys would be a reasonable future enhancement, but it does not represent a deficiency in the database as it stands today.

Feedback Point 2: SQL Injection via PHP

The feedback noted that SQL injection is a known risk when using PHP to interact with a database, and recommended the use of parameterized queries as a preventive measure. We agree that this is an important best practice for any web facing application and appreciate the reminder.

It is worth noting that this feedback pertains to the application layer, not to the database structure itself. Our database schema, constraints, and queries are not the source of this risk. In any future development of a PHP front end for this project, we would implement PDO prepared statements to ensure user input is never concatenated directly into SQL query strings. This is acknowledged as sound security advice going forward.

Feedback Point 3: Overall Design

Reviewers noted that our team demonstrated strong foresight in designing the database to accommodate future programming languages, and that our presentation showed genuine understanding of the material. This feedback required no revision. It confirmed that our design decisions were well founded.

Part II: Implementation of Revisions

3. Revised Database Structure

No structural changes were made to the database. As documented in Part I, the feedback received did not identify any errors, constraint violations, or normalization issues in our current schema. Our database remains in Third Normal Form, which we achieved by identifying and correcting a transitive dependency prior to our Phase II submission.

Specifically, the version attribute was moved from the implementation table to the language table, resolving the transitive dependency: implementation_id to language_id to version. Since version is a property of a language rather than of any individual implementation row, it correctly belongs in the language table where it depends directly on language_id. This change was fully reflected in our three lookup views, vw_function_lookup, vw_operator_lookup, and vw_structure_lookup, which were updated to retrieve version via a JOIN on the language table.

The database structure submitted in Phase II is correct, normalized, and requires no further revision based on the feedback received.

4. Revised SQL Queries

No changes were made to our SQL queries. The queries submitted in Phase II correctly reflect the normalized schema and return accurate results. The views vw_function_lookup, vw_operator_lookup, vw_structure_lookup, vw_function_quick_reference, and vw_operator_category_summary all function as intended and were not flagged for errors in the feedback.

The feedback we received focused on future proofing and security practices rather than on correcting existing query logic. Our current queries accurately join through the language table for version data, enforce foreign key relationships across all junction tables, and return consistent results across all views. We are confident the query layer of our database is sound and does not require revision at this stage.